A Label-Replication Artefact, a Measured Leakage Ladder and a Null-Anchored Benchmark for PPG-Based Blood Glucose Estimation

Gaurav Verma

***Abstract*—Reported accuracies for blood-glucose estimation from a single photoplethysmography (PPG) channel often rival regulated glucose monitors, and two concurrent studies have now shown that such numbers largely disappear under subject-wise evaluation. This paper adds what those studies do not provide. First, we characterise a *label-replication artefact*, for which we found no prior report, in a public Kaggle derivative of the Kermani–Esmaeili PPG–glucose recordings: within each subject the glucose label cycles row by row through the subject's reference values, so every window inherits the subject mean and a per-subject oracle reaches MAE 0.03 mg/dL. This is a label-construction leak that subject-wise splitting cannot detect; we release a full-file detector for it. Second, we measure a *leakage ladder*: the same 1-D CNN scores MAE 10.4±0.2 mg/dL ($r = 0.66 \pm 0.05$) under a random-window split but 13.5±0.5 and 13.4±0.5 once recordings or subjects are separated, against 14.7 for the train-mean null on the raw recordings (67 recordings, 23 subjects): on this cohort window-neighbour leakage, not identity leakage, inflates the result. Third, we release PPG-BENCH, a benchmark harness with machine-checked no-leak split protocols, frozen folds, null, classical, published-architecture and frozen pretrained-encoder rows (PaPaGei-S, Pulse-PPG, AnyPPG), subject-cluster bootstrap confidence intervals, positive controls and clinical anchoring (Parkes zones, hypo/hyper event sensitivity, reference-noise floors). On the PhysioNet BIG IDEAs wearable cohort (16 subjects, 8664 held-out CGM readings, chronological split with a 1-h guard gap) a population model is significantly worse than each subject's own mean (1-D CNN ΔMAE +1.7 mg/dL, 95% CI [+0.5, +3.1]); a personalised arm that gives the models the same subject information as the null reaches 15.0±0.0 mg/dL against 15.3; frozen probes of three PPG foundation models (PaPaGei-S, Pulse-PPG, AnyPPG) do not materially improve on it, the best (Pulse-PPG) reaching a paired difference of -0.3 mg/dL (95% CI [-0.5, -0.02]), far below the reference-noise floor of 10.5 mg/dL; and no PPG model detects any of the 68 hypo- or 150 hyperglycaemic readings. Every benchmark number in this paper is generated from the released prediction files.**

***Index Terms*—photoplethysmography, blood glucose, non-invasive sensing, data leakage, benchmark, subject-wise cross-validation, null model, Clarke and Parkes error grids, reproducibility.**

I. Introduction

An optical wearable that reports blood glucose would remove the fingerprick burden from the roughly 589 million adults living with diabetes [1]. Photoplethysmography (PPG) is the obvious candidate signal: it is already captured continuously by consumer wristbands, and glucose affects blood viscosity, vascular compliance and erythrocyte deformability, which leave traces in pulse morphology [2]. Published PPG-only deep regressors report mean absolute errors (MAE) of 3–12 mg/dL [3]–[5] and MARD of 18% [6], close to the accuracy of regulated continuous glucose monitors (CGM); the U.S. FDA has nonetheless warned consumers that no smartwatch or ring measures glucose non-invasively [7].

Two studies published while this work was under way explain much of that gap. Ramesh *et al.* [8] re-implemented five published PPG-glucose methods on VitalDB (4 626 participants) and found near-zero or negative R^2, no better than a mean predictor, once splits were participant-aware. Seyedebrahimi *et al.* [9] showed on the BIG IDEAs wrist cohort that adding PPG, electrodermal activity, temperature and acceleration does not improve on a cohort-mean or time-of-day predictor under subject-grouped cross-validation. Ramesh *et al.* further report that more than 90% of the mean predictor's outputs fall in Clarke zones A+B, so zone statistics alone do not separate a model from its null. The message that random splits inflate PPG-glucose results is therefore no longer new, and this paper does not claim it.

What this paper adds is the material that makes such a message actionable:

1) **A label-replication artefact** in a public Kaggle derivative of the Kermani–Esmaeili recordings [10], [11] (Sec. III-B). Within each subject the label cycles row by row through that subject's reference values; every multi-row window therefore carries the subject mean, and a per-subject oracle scores MAE 0.03 mg/dL. In the taxonomy of Kapoor and Narayanan [12] the nearest type is duplication, but the mechanism is replication of the label column rather than of rows, and it survives a perfectly subject-wise split. We release a full-file detector. An earlier, unpublished draft of this work trained on that file; its numbers are not carried forward.
2) **A measured leakage ladder** (Sec. V-A): the same models evaluated under random-window, random-recording, subject-wise and derivative-file protocols on one dataset, and under random-reading, chronological and guarded-chronological protocols on a wearable CGM cohort. Ramesh *et al.* [8] already compare random, participant-aware and leave-some-participants-out splits with a mean baseline on VitalDB, and other reports give single before/after pairs [13]–[15]; the ladder here adds the rungs those studies lack: overlapping-window ver-

G. Verma is with the Department of Electronics and Communication Engineering, [Institution Name] (e-mail: [author.email@institution.edu]). Code, frozen folds, integrity tests and all out-of-fold predictions: https://github.com/Namangarg484/IEEE-Gaurav-Verma.

sus recording separation on 10-s recordings, a label-construction rung on the derivative file, leave-one-subject-out on the 23-subject cohort itself, and random-reading versus guarded-chronological splits on a CGM cohort.

3) **PPGBENCH, a null-anchored benchmark harness** (Sec. IV): named split protocols with machine-checked no-leak assertions and frozen fold files; null, classical, published-architecture and frozen pretrained-encoder rows under one protocol; subject-cluster bootstrap intervals with a pre-specified primary comparison and a stated minimum detectable effect; positive and negative controls; and clinical anchoring (Parkes zones, hypo/hyper event sensitivity, reference-noise floors). Ramesh *et al.* [8] release a reproducible evaluation pipeline for five re-implemented methods on VitalDB; PPGBENCH differs in asserting each protocol's no-leak guarantee in code, freezing folds, and adding subject-cluster intervals, controls, within-subject nulls and pretrained-encoder rows.
4) **A fair within-subject test** on BIG IDEAs (Sec. V-C): a chronological split with a guard gap against each subject's own mean, time-of-day and persistence nulls, and a personalised arm that gives the models the same subject information as the null, so that the remaining gap is attributable to the PPG channel rather than to the design.

II. Related Work

A. Reported PPG-glucose accuracy on public data

Table I lists results reported on the same 23-subject Kermani–Esmaeili recordings used here, with the split each study describes. The spread (MAE 3–16 mg/dL) is explained by the split rather than by the model: the sub-10 mg/dL results come from protocols that place windows or augmented copies of one recording on both sides of the split, and every study that evaluates on unseen subjects, including Syamsul and Kim's held-out subjects without a null row, lands within about 3 mg/dL of the train-mean null (14.7). On larger data, multi-view fusion transformers [5] and multi-view attention networks [6] published in this journal report 88% of predictions in Clarke zone A and 99% in zones A+B, respectively, without a null-model row; Ramesh *et al.* [8] show that zone A+B fractions above 90% are reached by the mean predictor, so a zone fraction without a null row is uninterpretable. Studies that do obtain subject-wise accuracy use per-user calibration readings [16], [17] or minute-scale multimodal features with food logs [18].

TABLE I
PUBLISHED RESULTS ON THE KERMANI–ESMAEILI RECORDINGS (23 SUBJECTS, 67 FINGERSTICK REFERENCES) AND THE NULL-ANCHORED RESULT OF THIS PAPER. MAE IN MG/DL.

Study	Model	Split	MAE
Soliman 2024 [3]	CNN–GRU	augment, then shuffle	2.96
Kermani & Esmaeili 2025 [19]	SDF-MFCC + SVR	5-fold, level unstated	4.0†
Syamsul & Kim 2026 [4]	attention CNN–GRU	subject-level (stated)	11.59
Syamsul & Kim 2026 [4]	1-D CNN baseline	subject-level (stated)	13.45
Zeynali 2025 [20]	ResNet-34, VitalDB-trained	external test	14.8
Ramesh 2026 [8]	train-mean null, VitalDB	external test	16.0
This paper	train-mean null	LOSO, recording level	14.7
This paper	1-D CNN, 3 seeds	LOSO, recording level	13.4±0.5

†reported together with RMSE 26.0 and MARD 0.195%, which are mutually inconsistent.

B. Leakage and evaluation in physiological machine learning

Record-wise splits of autocorrelated signals are the best-documented leak in wearable regression: segment- and record-level splits inflate the Pearson correlation of PPG-based mean-BP estimation by 0.56 and 0.40 relative to a subject-level split [13], and the error standard deviation of a published PPG-to-BP network rises from 8 (overlapping windows) through 16 (record-wise) to 23 mmHg (subject-wise) [14]; EEG decoding accuracy drops from 99.8% to 53% [15]; and 42% of wearable blood-pressure studies in a recent meta-analysis use record-level splits [21]. Guidance exists on approximating the use case in the split [22], on identity confounding and subject-level permutation tests [23], on the width of error bars at small n [24] and on the distributional bias of leave-one-out estimates [25], and reporting standards now ask for the generalisation level to be stated [26], [27]. Kapoor and Narayanan [12] classify leaks into eight types; the artefact reported here is closest to their duplicates type but arises during the construction of the label column of a public table, which none of the types names explicitly.

C. Multi-view PPG representations and pretrained encoders

Wavelet scalograms and Takens delay embeddings expose time–frequency and phase-space structure [28], and PointNet [29] encodes the resulting point clouds. The network evaluated here fuses these views with a learnable-query attention [30]. Self-supervised PPG encoders trained on 10^4–10^5 hours of data (PaPaGei [31], Pulse-PPG [32], AnyPPG [33]) now exist; linear probes of all three on emergency-department PPG give glucose R^2 of 0.03–0.07 (0.08 for the PPG-LM encoder itself) [34]. We probe all three under the benchmark's protocols, so that the null is tested against three openly available pretrained representations and not only against a small CNN.

III. Datasets

A. Kermani–Esmaeili finger-PPG recordings

The dataset of Kermani and Esmaeili [10], [19] (Mendeley Data, v3/v4; the `.csv` layout of v3 is read here) contains 67 recordings from 23 adults (15 male, 8 female, 22–61 years from the label files). Each recording is 21 900 samples of an index-finger pulse sensor (550-nm green LED, APDS-9008 photodiode), i.e. 10 s at 2175 Hz, with one concurrent capillary reference from an Accu-Chek Active meter, plus age, sex, height and weight. Diabetes status is not stated by the dataset

authors. Subjects contributed one to seven recordings at different times (12 of 23 contributed at most two), so the raw data carry within-subject variation: subject 01 spans 99–136 mg/dL over seven recordings (Fig. 1b). The reference ranges from 88 to 183 mg/dL (mean 115.0 ± 18.6); 2 recordings exceed 180 mg/dL and none is below 70. The `Labels` folder holds 68 files because `Total.csv` concatenates the 67 per-recording labels. The raw pulsatile amplitude is 2–6 ADC counts on a ∼512-count offset, so band-pass filtering precedes every representation.

B. The Kaggle derivative and its label-replication artefact

A derivative of these recordings is distributed on Kaggle as "PPG signal with Blood sugar level data" [11]: a single `clean-dataset.csv` (844 946 rows, 46.8 MB) with columns `PPG_Signal`, `Patient_Id`, `Heart_Rate`, `Systolic_Peak`, `Diastolic_Peak`, `Pulse_Area`, `Gender`, `Age`, `Glucose_level`, `Height`, `Weight` and `pl`, without Mendeley provenance. It was the training source of the first version of this work. Inspecting it row by row (Fig. 1a) shows that within each subject `Glucose_level` cycles through the subject's k reference values with period k while `PPG_Signal` advances one sample per row and `pl` counts $1, \ldots, k$: the table is a cross-join of each subject's signal with its label list. Any window of $W \gg k$ rows therefore has a mean label equal to the subject mean, and the within-subject variation visible in the raw recordings is removed. Two consequences follow. A per-subject oracle that predicts each window with the mean of its own subject's labels achieves MAE 0.03 mg/dL with $r = 1.00$ and 100.0% zone A (Sec. V-A), which an earlier, unpublished draft of this work had reported as a "dataset property". And a model trained on this file only ever learns to map PPG morphology to a subject constant, so a correct subject-level split evaluates between-subject baseline estimation on a few held-out subjects whose means happen to lie close together; that draft's MAE of 6.89 mg/dL and 98.9% zone A on four held-out subjects are artefacts of this file and are not carried forward. We found no peer-reviewed paper that uses this derivative; the published results on this cohort (Table I) use the raw Mendeley files. The released integrity test streams the entire file and asserts the period-k cycling for every subject, and asserts the one-label-per-recording property of the raw data.

C. PhysioNet BIG IDEAs wearable cohort

The BIG IDEAs Lab Glycemic Variability and Wearable Device dataset (PhysioNet v1.1.3) [18], [35] pairs an Empatica E4 wristband (blood-volume pulse, BVP, 64 Hz) with a Dexcom G6 CGM (one reading per 5 min) in 16 adults (9 female, 7 male; HbA1c 5.3–6.4%, i.e. high-normal to prediabetic; `Demographics.csv`) over 8–10 days of free living. Bent *et al.* [18] reported on this cohort that personalised gradient-boosted models using minute-scale heart-rate, electrodermal, temperature and accelerometer features together with food logs predict interstitial glucose better than naive mean and median baselines; we test the PPG channel alone. Each

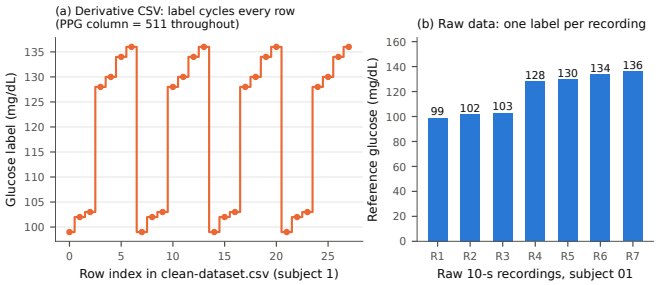


Fig. 1. The label-replication artefact. (a) In the Kaggle derivative the glucose label of subject 1 cycles through the subject's seven reference values on consecutive rows while the PPG column holds a single sample value; any multi-row window averages to the subject mean. (b) The raw recordings of the same subject carry one reference value each and span 99–136 mg/dL.

subject's BVP file (≈1.4 GB) is streamed once and the two minutes of BVP around every CGM timestamp are kept. For modelling we use the last two gap-free 5-s windows within the 60 s preceding each reading; windows after the reading are never used, and the interstitial lag of the reference (3–4 min for the G6 [36]) is not compensated. The resulting cohort comprises 8664 held-out and 28867 total CGM readings with glucose 46–261 mg/dL (mean 116.5 ± 23.7); the held-out period contains 68 readings below 70 and 150 above 180 mg/dL.

D. STUBA ePPG multi-wavelength recordings

As a third cohort we use the ePPG dataset [37] (Zenodo, CC0): six-channel finger PPG (530, 655 and 940 nm at two sites) at 100 Hz in 30-s records with a Beurer GL49 fingerstick reference, 24 subjects (20 male, 4 female, 22–48 years; 3 with type-1 diabetes) and 125 records with glucose 40–304 mg/dL, including 34 hypo- and 18 hyperglycaemic references. Two diabetic subjects contributed 32 and 53 records; 17 subjects contributed at most two. The 940-nm channel (IR1) is used by default. It is evaluated with the same leave-one-subject-out protocol as the Kermani–Esmaeili data and provides a replication on a second device, a per-wavelength ablation, and the only cohort with hypoglycaemic references.

IV. Benchmark Design

A. Common representation

Every source is mapped to one representation: band-pass 0.5–8 Hz (fourth-order Butterworth, zero phase) at the native rate, polyphase resampling to $f_s = 64$ Hz, 5-s windows of $W = 320$ samples, z-scored per window. Kermani–Esmaeili recordings yield three windows each with 50% overlap (201); BIG IDEAs readings yield two windows. Each window feeds three views: the waveform (1×320), a Mexican-hat wavelet scalogram over 64 scales resized to 64×64 and min–max scaled, and a Takens embedding $\mathbf{p}_i = [\hat{x}[i], \hat{x}[i+5], \hat{x}[i+10]]$ with 256 points. Views are computed once and cached, so a fold only indexes tensors. This representation deliberately discards DC/perfusion amplitude, vasomotion and heart-rate variability; Sec. V-G reports how the conclusions change when they are retained.

B. Split protocols with machine-checked no-leak guarantees

PPGBENCH names every protocol and asserts its guarantee before any model is trained: `window_random` (overlapping windows of one recording may fall on both sides), `record_random` (recordings separated, subjects shared), `subject_kfold` and `subject_loso` (subjects separated), `chronological` (per subject, the first 70% of readings by time train, the rest test, with a guard gap of g seconds removed from the training side) and `personal` (chronological, one model per subject). Folds are frozen to JSON files shipped with the code, and a `leak_report` counts shared windows, recordings, subjects and time overlaps per fold. The primary protocols are `subject_loso` for the fingerstick cohorts and `chronological` with $g = 3600$ s for BIG IDEAs; the others are rungs of the ladder.

C. Models

Nulls. Train-mean (fingerstick cohorts); each subject's own training-period mean, a time-of-day ridge (sin/cos of hour plus subject offset) and persistence (the subject's last training-period value) (BIG IDEAs). **Classical.** Ridge and gradient boosting on nine hand-crafted PPG features (Welch heart rate, three relative band powers, log power, skewness, kurtosis, mean absolute derivative, zero-crossing rate of the derivative), with and without demographics. **Networks.** The 1D–2D–3D cross-attention fusion network of our earlier draft (Fig. 2; 548 801 parameters) and its single-view branches; re-implemented ResNet-1D and CNN–LSTM regressors as used in the PPG-glucose literature [4], [20]. **Frozen pretrained encoders.** PaPaGei-S [31] (5.8 M parameters, 125 Hz), Pulse-PPG [32] (28.5 M, 50 Hz) and AnyPPG [33] (4.0 M, 125 Hz), each with the authors' released weights, frozen. Every segment is band-passed 0.5–12 Hz (fourth-order Chebyshev type II, the band-pass stage of PaPaGei's pyPPG preprocessing, without its 50-ms smoothing) at the native rate, resampled to the encoder's rate and z-scored. One pipeline is used for all three encoders; it differs from AnyPPG's pretraining filter (0.5–8 Hz Butterworth) and from Pulse-PPG's (unfiltered, person-normalised 4-min windows), and Pulse-PPG therefore receives far shorter inputs than it was pretrained on. PaPaGei-S and AnyPPG receive 10-s segments: the whole Kermani–Esmaeili recording, three per 30-s STUBA record, and the two most recent gap-free 10-s segments of the 60 s before each BIG IDEAs reading; Pulse-PPG, which pools over time, receives the whole record (10 s Kermani–Esmaeili, 30 s STUBA) or, on BIG IDEAs, those two segments concatenated (not necessarily contiguous; 10 s when only one passes). Segment embeddings (512-d) are averaged per reference measurement and a ridge regression (penalty by internal cross-validation, features standardised on the training fold) is fitted under the same protocol as every other row. **Personalised arm.** On BIG IDEAs, networks and ridge are additionally trained on the residual $y - \bar{y}_s^{\mathrm{train}}$ and the subject's training mean is added back, so that they hold exactly the information the subject-mean null holds.

D. Training

Every network in a fold is trained from scratch for a fixed budget (40 epochs on the fingerstick cohorts, 15 on BIG IDEAs) with AdamW (learning rate 10^{-3}, weight decay 10^{-4}), a one-cycle schedule, batch size 16 (64 on BIG IDEAs), gradient clipping at norm 1 and a smooth-L1 loss on the target standardised with the *training* statistics of that fold. There is no early stopping and no selection on the held-out subject. One shared configuration is used for all networks and no model was tuned individually; Sec. V-G argues from the paired-interval half-width why this does not change the conclusions.

E. Metrics and statistics

Window predictions are averaged per reference measurement, and every metric is computed on the out-of-fold measurement-level predictions: MAE, RMSE, MARD, Pearson r, within-subject r (median over subjects), and Clarke and Parkes/Consensus zones [38], [39]. Because measurements within a subject are correlated, uncertainty is estimated with a cluster bootstrap that resamples subjects with replacement (5 000 draws): for each model we report the 95% percentile interval of its MAE and of its *paired* MAE difference against the null, computed on the same resampled subjects, and a two-sided sign test on the per-subject paired MAE differences (reported as the number of subjects for which the model is worse than the null). Per-measurement tests are not used because measurements within a subject are not independent. The primary comparison is pre-specified as *1-D CNN versus the null*; all other rows are exploratory. The half-width of the paired interval approximates the smallest difference that would exclude zero at the 95% level (the percentile interval is not symmetric; about 1.5 mg/dL for the fingerstick primary comparison, 1.3 for the BIG IDEAs population comparison and 0.5 for the personalised arm); we use it as a design-sensitivity bound, not as a power calculation. Networks are run with several seeds; MAE, RMSE and r are reported as mean $\pm$ standard deviation over seeds, while the paired difference and its interval are computed on the seed-ensemble prediction (the mean of the seeds' predictions per measurement), whose MAE is given alongside. Under leave-one-subject-out, a predictor that regresses toward its training mean acquires a negative pooled correlation with the held-out labels (the null has $r = -0.59$ here), a known distributional bias of leave-one-out estimates [25]; we therefore report the null's r rather than hide it and add a 5-fold subject-grouped sensitivity row.

F. Controls

Through the identical windows, views, model and protocol we regress quantities that must be recoverable (the window's own heart rate; subject age) and quantities that must not be (glucose labels permuted across recordings; white-noise input with the true labels). A benchmark whose positive controls fail cannot support a negative result.

G. Clinical anchoring

Three quantities are reported next to every null: the *reference-noise floor*, i.e. the MAE and MARD that a perfect

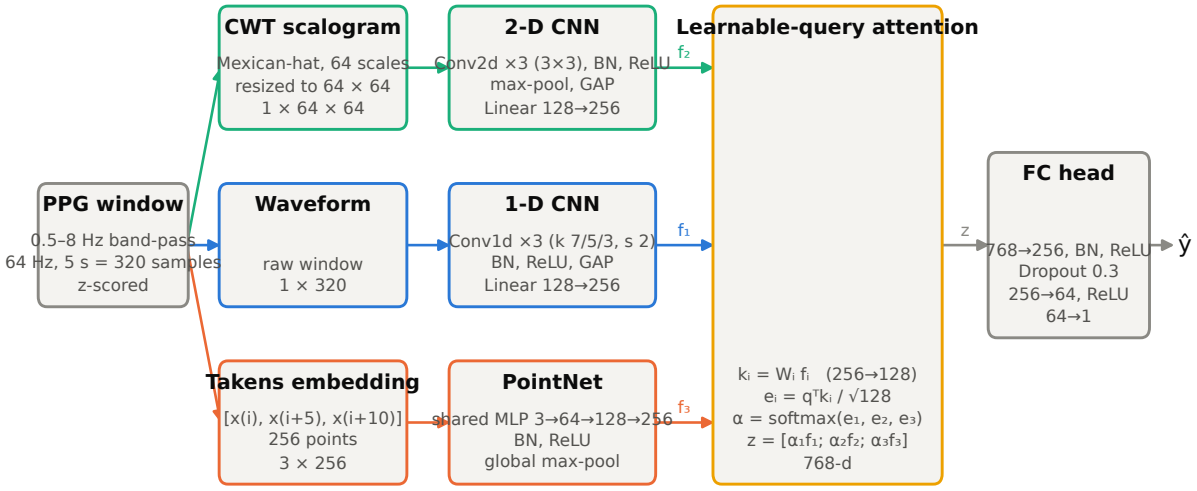


Fig. 2. The 1D–2D–3D fusion network evaluated as one row of the benchmark. One PPG window is presented as a waveform, a wavelet scalogram and a Takens phase-space point cloud; three equal-width encoders feed a learnable-query attention that weights the branches before a regression head.

estimator would score against a reference carrying the device's own error (ISO 15197:2013 meter limits for the fingerstick cohorts; MARD 9% [40] for the CGM cohort); event sensitivity and specificity for hypoglycaemia (< 70 mg/dL) and hyperglycaemia (> 180 mg/dL); and time-in-range agreement. The intended use case of a spot-check estimator (calibration-free, no history) differs from that of a CGM replacement (history available), and the null that applies to each differs accordingly: train-mean for the former, subject-mean, time-of-day and persistence for the latter.

H. Availability

The release comprises (i) the PPGBENCH package: named split protocols whose leak assertion runs before any training, a `leak_report` per fold, the full-file label-cycling detector, metrics, cluster-bootstrap and report code, and a command-line entry point; (ii) frozen fold files (four Kermani–Esmaeili, five BIG IDEAs and two STUBA protocols) with fingerprints, so that a third-party model is scored on exactly these folds or is flagged; (iii) one out-of-fold prediction file per model, protocol and seed, from which every number in this paper is regenerated; (iv) the loaders for the three public cohorts (none of which is redistributed) and the integrity test suite; and (v) the TRIPOD+AI checklist and model information sheet of the supplement.

V. Results

A. Leakage ladder

Fig. 3 and Table II evaluate the same 1-D CNN under protocols that leak progressively less. On the Kermani–Esmaeili recordings a random split over 5-s windows places overlapping windows of the same 10-s recording on both sides of the split and yields MAE 10.4±0.2 mg/dL with $r = 0.66 \pm 0.05$ and zone A 91.0±1.2%: numbers that read as a working sensor. Separating recordings but sharing subjects gives 13.5±0.5; separating subjects gives 13.4±0.5. The fusion network follows the same ladder (10.7, 14.9 and 14.8±0.3 mg/dL, with $r = 0.71$ under the window split), so the inflation is a property of the split, not of the architecture. The leaky subject-mean predictor, which simply knows the test subject's mean, scores 17.8 under the recording split, *worse* than the train-mean, because 12 subjects contribute at most two recordings; on this dataset window-neighbour leakage, not identity leakage, is the inflating mechanism. The derivative-file rungs show the artefact at work: on windows cut from the Kaggle file, any predictor that can identify the subject scores near zero, because the labels are subject constants: under a random window split the leaky subject-mean reaches 0.03 mg/dL with $r = 1.00$ and 100.0% zone A, the "perfect sensor" that a model with access to subject identity (or a per-subject calibration) would report on this file. The 1-D CNN itself scores 9.5 under that split and 10.2 under a correct subject split, against a null of 10.2: the derivative's waveform column carries too little identity for the network to exploit the leak in a fixed budget, but the null itself is lower than on the raw recordings because the collapsed labels have less variance, and a four-subject validation set whose means lie close together is the regime that produced the first draft's 6.89 mg/dL. On BIG IDEAs the ladder is flat: a random split over CGM readings gives 16.1 (scored on all 28867 pooled out-of-fold readings), a chronological split without a guard gap 16.9, and the guarded split 16.9, against subject-mean nulls of 15.0, 15.2 and 15.3. Reading-level leakage lowers the model's error by about 0.9 mg/dL but leaves it above the null, and the guard gap changes the chronological result by less than 0.01 mg/dL at this window length: no rung lets a PPG model beat the null on this cohort.

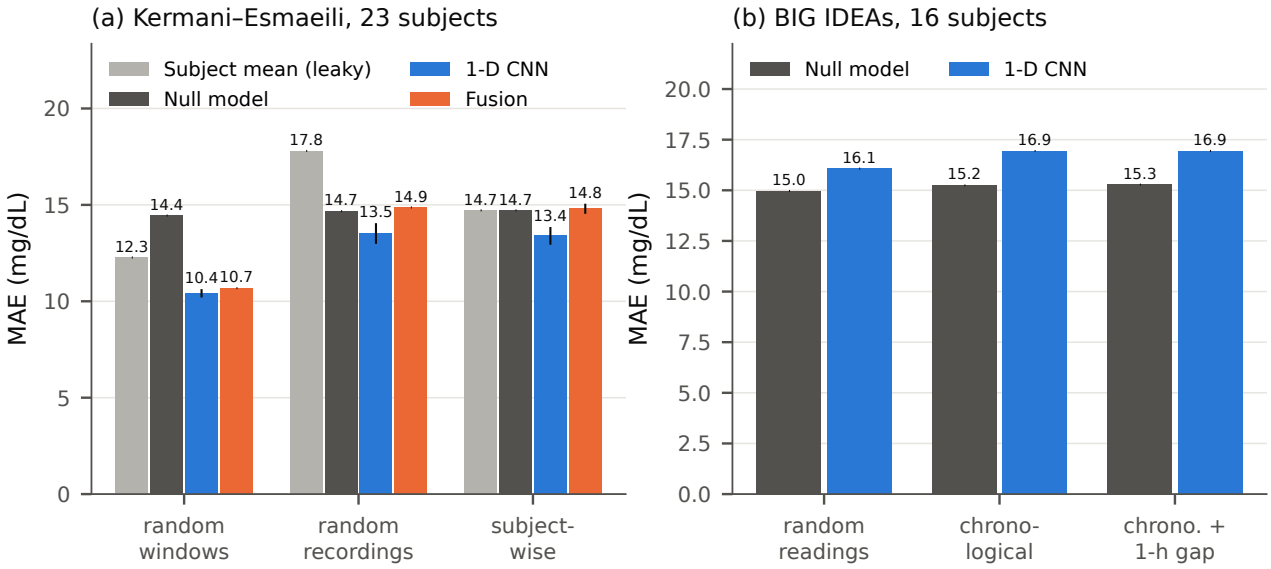


Fig. 3. Leakage ladder: MAE of the same models under progressively less leaky protocols. (a) Kermani–Esmaeili recordings: random windows, random recordings (subjects shared), subject-wise. (b) BIG IDEAs: random CGM readings, chronological without a guard gap, chronological with a 1-h gap. "Subject mean (leaky)" predicts each test window with the test subject's own training mean; on BIG IDEAs the null is that same subject mean. Error bars: std over three seeds (1-D CNN, all Kermani–Esmaeili rungs; fusion, subject-wise rung, which is the experiment of Table III); all other bars are single runs.

TABLE II
LEAKAGE LADDER, 1-D CNN AND NULLS; MAE IN MG/DL. NULL: TRAIN MEAN FOR THE KERMANI–ESMAEILI RECORDINGS AND THE KAGGLE DERIVATIVE; EACH SUBJECT'S OWN TRAINING-PERIOD MEAN FOR BIG IDEAS, WHICH COINCIDES WITH THE LEAKY SUBJECT MEAN. K–E WINDOW, RECORDING AND LOSO RUNGS: THREE SEEDS (MEAN ± STD, THE LOSO RUNG BEING THE EXPERIMENT OF TABLE III); ALL OTHER RUNGS: ONE SEED. THE BIG IDEAS RANDOM-READINGS RUNG IS SCORED ON ALL 28867 POOLED OUT-OF-FOLD READINGS, THE CHRONOLOGICAL RUNGS ON THE 8664 HELD-OUT READINGS.

Cohort	Protocol	Null	Leaky subj. mean	1-D CNN
K–E	random windows	14.4	12.3	10.4±0.2
	random recordings	14.7	17.8	13.5±0.5
	subject-wise (LOSO)	14.7	–	13.4±0.5
Kaggle deriv.	random windows	9.7	0.03	9.5
	subject-wise (LOSO)	10.2	–	10.2
BIG IDEAs	random readings	15.0	15.0	16.1
	chronological, no gap	15.2	–	16.9
	chronological, 1-h gap	15.3	–	16.9

B. Cross-subject estimation on the Kermani–Esmaeili recordings

Table III and Fig. 4 give the leave-one-subject-out results with subject-cluster intervals. The train-mean null scores MAE 14.7 mg/dL. Ridge on PPG features (14.6) or on demographics (15.6) and gradient boosting on PPG features (16.6; Δ +1.9, CI [+0.2, +3.5], i.e. worse) or on PPG plus demographics (15.5) do not improve on it. The pre-specified primary comparison gives a 1-D CNN MAE of 13.4±0.5 mg/dL over seeds (13.3 for the seed-ensemble prediction), a paired difference of -1.4 mg/dL (95% CI [-3.0, -0.03]) and $r = 0.32 \pm 0.02$ (CI [+0.17, +0.50]); the upper limit of the interval is -0.03 mg/dL, the model is worse than the null for 8 of 23 subjects (sign test $p = 0.210$), and the advantage is nominally significant for 1 of 3 seeds taken individually, so we report it as weak evidence. Its correlation is partly leverage: dropping the 4 recordings above 140 mg/dL reduces r to +0.27. Under a 5-fold subject-grouped split, which reduces the leave-one-out distributional bias, the 1-D CNN scores 14.4±0.2 mg/dL with $r = 0.19 \pm 0.03$ against a null of 14.6 ($r = -0.24$): the null's negative correlation is the protocol's, not the model's, and the 1-D CNN's advantage shrinks to within noise, so its correlation should be read as 0.2–0.3 depending on the protocol rather than as a stable signal. The 2-D scalogram branch (13.9±0.2), the 3-D phase-space branch (14.0±0.1), ResNet-1D (15.1±0.4) and CNN–LSTM (13.5±0.2) have paired intervals that include zero (CNN–LSTM narrowly: -1.4, [-2.9, +0.2]), as does the full fusion network (14.8±0.3 over seeds, 14.7 as an ensemble; Δ -0.1, CI [-1.5, +1.2]). Under the shared configuration the fusion of the three views is therefore not better than its 1-D branch; Sec. V-G explains why no per-model tuning was attempted. Ridge probes on frozen embeddings of the three pretrained PPG encoders (Table III, bottom) score 14.7 (PaPaGei-S; Δ +0.0, CI [-0.9, +1.0]), 14.2 (Pulse-PPG; -0.5, [-1.7, +0.8]) and 14.9 mg/dL (AnyPPG; +0.2, [-0.2, +0.7]): encoders pretrained on 10^4–10^5 hours of PPG land on the null, like the networks trained from scratch.

C. Within-subject tracking on BIG IDEAs

Table IV and Fig. 5 report the guarded chronological split over 8664 held-out readings. Against a population model, each subject's own mean (MAE 15.3 mg/dL, MARD 12.8%) is *better* than every PPG model: the paired difference is +1.7 mg/dL (95% CI [+0.5, +3.1]) for the 1-D CNN and +1.4 ([+0.2, +2.9]) for the fusion network, and the 1-D CNN is worse than the subject mean for 13 of 16 subjects (two-sided sign test $p = 0.021$). The pooled r of 0.21 decomposes into a between-subject component (the models partly learn subject

TABLE III

LEAVE-ONE-SUBJECT-OUT RESULTS ON THE KERMANI–ESMAEILI RECORDINGS (67 RECORDINGS, 23 SUBJECTS, RECORDING LEVEL). MAE AND RMSE IN MG/DL; A AND A+B ARE CLARKE FRACTIONS IN %. NETWORKS: MAE, RMSE, r AND ZONES ARE MEAN ± STD OVER 3 SEEDS; Δ AND ITS 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL ARE COMPUTED ON THE SEED-ENSEMBLE PREDICTION (MEAN OF THE SEEDS' PREDICTIONS PER RECORDING), WHOSE MAE IS GIVEN IN PARENTHESES, AGAINST THE TRAIN-MEAN NULL. THE PRIMARY COMPARISON IS MARKED ⋆; OTHER ROWS ARE EXPLORATORY.

	Model	MAE	RMSE	r	A	A+B	Δ vs. null [95% CI]
baseline	Train mean (null)	14.7	18.9	-0.59	83.6	100.0	–
	Reference-noise floor (ISO 15197)	7.1	–	–	–	–	–
	Ridge, demographics	15.6	19.7	-0.17	86.6	100.0	+0.9 [-0.3, +2.3]
	Ridge, PPG features	14.6	18.9	-0.03	86.6	100.0	-0.1 [-0.6, +0.6]
	Gradient boosting, PPG + demogr.	15.5	20.6	-0.05	80.6	100.0	+0.8 [-1.3, +3.0]
network	1-D CNN ⋆	13.4±0.5 (13.3)	18.0±0.2	0.32±0.02	80.6±1.2	100.0±0.0	-1.4 [-3.0, -0.03]
	2-D CNN (CWT)	13.9±0.2 (13.9)	18.5±0.1	0.22±0.01	84.1±0.7	100.0±0.0	-0.8 [-2.5, +0.7]
	PointNet (Takens)	14.0±0.1 (14.0)	18.6±0.1	0.14±0.03	88.1±0.0	100.0±0.0	-0.7 [-1.8, +0.4]
	ResNet-1D [20]	15.1±0.4 (15.0)	19.2±0.5	0.19±0.05	80.6±1.2	100.0±0.0	+0.3 [-1.4, +2.0]
	CNN–LSTM [4]	13.5±0.2 (13.3)	18.7±0.3	0.19±0.02	85.6±0.7	100.0±0.0	-1.4 [-2.9, +0.2]
	1D+2D+3D cross-attention	14.8±0.3 (14.7)	19.3±0.4	0.13±0.05	78.6±0.7	100.0±0.0	-0.1 [-1.5, +1.2]
probe	PaPaGei-S + ridge [31]	14.7	19.1	-0.01	83.6	100.0	+0.0 [-0.9, +1.0]
	Pulse-PPG + ridge [32]	14.2	18.5	0.14	80.6	100.0	-0.5 [-1.7, +0.8]
	AnyPPG + ridge [33]	14.9	19.2	-0.33	85.1	100.0	+0.2 [-0.2, +0.7]

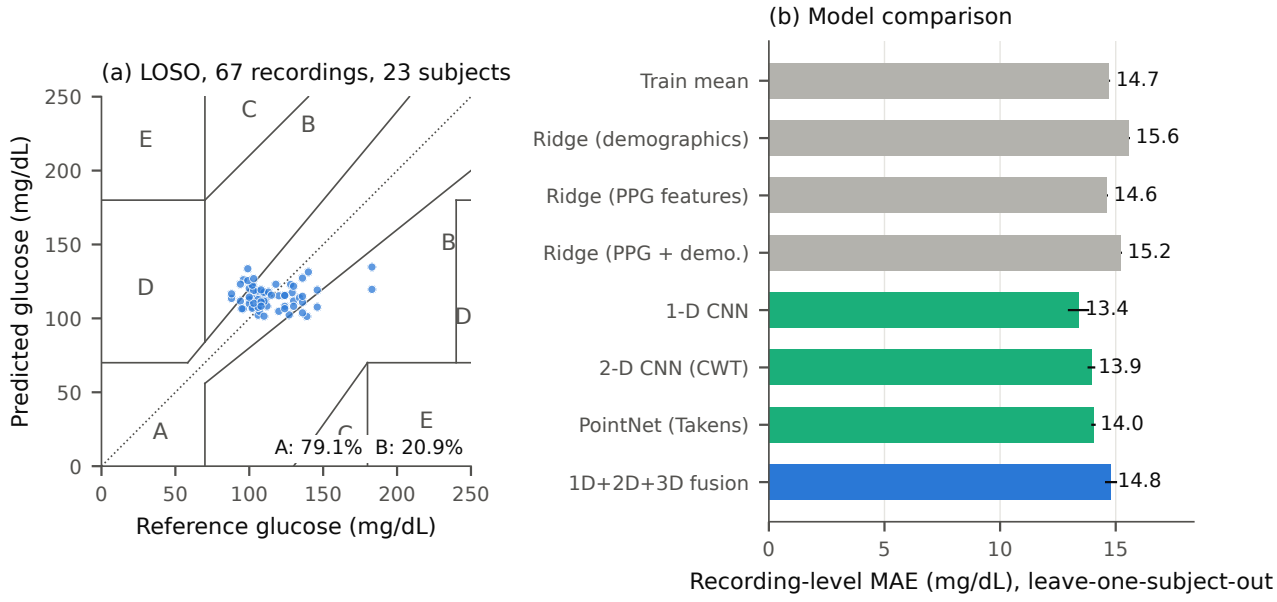


Fig. 4. Leave-one-subject-out results on the Kermani–Esmaeili recordings. (a) Clarke error grid of the fusion network's out-of-fold recording-level predictions: the predictions occupy a narrow band around the training mean. (b) Recording-level MAE of every model; grey bars are baselines that do not see the PPG waveform through a network.

offsets) and a within-subject median r of 0.04: the networks do not track excursions. The time-of-day null scores 15.3 and persistence 31.3.

The comparison above handicaps the models, which never see subject identity. The personalised arm removes that handicap: trained on the residual to the subject's training mean, ridge scores 15.1, the 1-D CNN 15.0±0.0 (Δ vs. subject mean -0.2, CI [-0.7, +0.2]) and the fusion network 15.1 (-0.2, [-0.7, +0.3]). The three seeds of the residual 1-D CNN agree to within 0.02 mg/dL, so the result is not seed noise; it is simply that the 1-D CNN adds nothing measurable to the subject's own mean.

Frozen pretrained encoders behave the same way. As population models they are worse than the subject mean (PaPaGei-S 17.2, Pulse-PPG 17.0, AnyPPG 16.8 mg/dL; all intervals above zero). With the residual target they reach 15.1, 15.0 and 15.1 mg/dL. Among the 40 exploratory paired comparisons in this paper (all listed in the supplement), the Pulse-PPG residual probe is the only one whose interval lies entirely below zero, i.e. in the model's favour (Δ -0.3 mg/dL, 95% CI [-0.5, -0.02]); 16 lie entirely above zero, i.e. the model is worse than its null. The effect is about 2% of the error and an order of magnitude below the reference's own noise floor (10.5 mg/dL), the model is worse than the subject mean for 4 of 16 subjects

(sign test $p = 0.077$), its median within-subject correlation is 0.07, and it would not survive any correction for the number of comparisons. We therefore read it as the largest benefit any frozen PPG encoder showed over a personal baseline on this cohort (within [-0.5, -0.02] mg/dL), not as tracking. The largest within-subject correlation of any model on BIG IDEAs, AnyPPG's residual probe ($r_w = 0.17$), has a cluster-bootstrap interval of [-0.03, +0.26] that includes zero. Fig. 5a shows the typical behaviour: the network output stays within a few mg/dL of the subject's mean while the CGM trace moves by 40–80 mg/dL after meals.

D. STUBA ePPG replication

The ePPG cohort is the hardest case for cross-subject estimation: two of its 3 type-1 diabetic subjects contribute all 34 hypoglycaemic and all 18 hyperglycaemic references, which no other subject resembles, so under leave-one-subject-out even the train-mean null scores MAE 43.5 mg/dL. Ridge on PPG features (44.4; Δ +0.9, CI [+0.3, +1.2]) and gradient boosting (48.3; +4.8, [+1.8, +9.2]) are significantly *worse* than the null, the 1-D CNN scores 49.7$\pm$1.3 (+6.1, [+2.3, +7.6]), the 2-D branch 48.6 (+5.1, [+2.7, +6.3]), the 3-D branch 43.2 (-0.3, [-1.3, +1.9]), the fusion network 47.5 (+4.0, [+1.3, +5.2]), and the frozen encoder probes 43.9 (PaPaGei-S), 44.6 (Pulse-PPG; Δ +1.1, CI [+0.1, +1.9]) and 43.5 (AnyPPG). Only the 3-D branch, PaPaGei-S (Δ +0.4, CI [-0.01, +1.7]) and AnyPPG (Δ +0.0, CI [-0.7, +1.1]) have intervals that include zero; nothing beats the null. Table V repeats the classical rows for each wavelength and site and for the six-channel average: no wavelength carries cross-subject glucose information that the others lack. None of the 34 hypoglycaemic references is detected by any model (Sec. V-F).

E. Controls

Table VI shows that the pipeline recovers what it should and nothing it should not. Heart rate, computed from the window's own spectral peak, is recovered with $r = 0.74 \pm 0.01$ and half the null's error (20.5$\pm$0.3 vs. 41.9$\pm$0.0 bpm; the ceiling is set by the noisy peak-picked target), and on BIG IDEAs with $r = 0.63$ (9.7 vs. 14.2 bpm). Permuted glucose labels (MAE 15.6$\pm$0.3 vs. null 14.6$\pm$0.0) and white-noise input (14.6$\pm$0.4 vs. 14.7$\pm$0.0) return to the null as they must. Age is not recovered ($r = -0.09 \pm 0.01$), which is informative in its own right: a single z-scored 5-s window carries no usable demographic signal under leave-one-subject-out, so the weak glucose correlation of the 1-D CNN cannot be an age proxy.

F. Clinical anchoring

The reference-noise floors are MAE 7.1 mg/dL (ISO 15197 meter, fingerstick cohort) and 10.5 mg/dL / MARD 9.0% (CGM cohort): the subject-mean null's MARD of 12.8% is within four points of the reference's own error, so the margin a PPG model would need to demonstrate clinical value is small in absolute terms but has to be shown against the subject mean, not against zero. None of the PPG models of Table IV, population or personalised, detects any of the 68 hypoglycaemic or 150 hyperglycaemic held-out readings, and all share the null's time-in-range agreement of 0.975. The only row with non-zero event sensitivity is the persistence null, which flags 48 hyperglycaemic readings by repeating a subject's last training value (specificity 0.88); that is a property of the subject's history, not of the PPG. Parkes zone A fractions are 82.6% for the null and 77.9% for the 1-D CNN.

G. Preprocessing and configuration sensitivity

The common representation could hide glucose information that lives in the DC level, in slow vasomotion or in heart-rate variability. Table VII repeats the BIG IDEAs personalised comparison (ridge and 1-D CNN on residual targets against the subject-mean null) with one window per reading of 5, 10, 30 and 60 s, with a 0.1–12 Hz band, with the per-subject amplitude retained instead of per-window z-scoring, and with log raw-window amplitude and, for windows of 30 s and longer, SDNN/RMSSD proxies from systolic-peak intervals added to the ridge features. No representation moves either model by more than 0.5 mg/dL from the null (paired intervals were not computed for these single-seed ablations): the best configuration (60-s windows, 0.1–12 Hz, amplitude and HRV) reaches 15.0 mg/dL for the 1-D CNN and 15.0 for ridge against the null's 15.3, i.e. a third of a mg/dL, and the median within-subject correlation does not exceed 0.1 in any configuration (largest: 0.10, ridge on 60-s windows with amplitude and HRV features). Longer windows help monotonically but by tenths of a mg/dL; the band and the amplitude make no difference. The negative result is therefore not an artefact of the 5-s, DC-stripped window. Hyper-parameters were shared across networks (one learning rate, budget and batch size) and no nested sweep was run. The personalised comparison's interval half-width (about 0.5 mg/dL) approximates the gain a tuned model would need before it could be distinguished from the subject mean in this design; the largest gains observed, -0.3 mg/dL for the Pulse-PPG probe and the 60-s 1-D CNN of Table VII (no paired interval computed), are of that order, so tuning could change a conclusion only for effects of this sub-mg/dL size, and the fusion-versus-1-D comparison is reported as an observation under the shared configuration rather than as a capacity claim.

VI. Discussion

What is new relative to the concurrent studies. Ramesh *et al.* [8] established that random splits inflate PPG-glucose results and that Clarke zone A+B fractions above 90% are reached by a mean predictor, and Seyedebrahimi *et al.* [9] that wrist signals add nothing to a cohort-mean or time-of-day predictor under subject-grouped cross-validation. This paper contributes the artefact that defeats even subject-wise splitting, a ladder that measures each inflation mechanism on the same data, a benchmark harness that, beyond the pipeline of [8], asserts every protocol's guarantee before training, and a within-subject test that gives the models the same subject information as the null. With those in place the conclusion is narrower and more useful: on 23+16+24 subjects and under every representation and architecture tested, the information a

TABLE IV

WITHIN-SUBJECT CHRONOLOGICAL SPLIT WITH 1-H GUARD GAP ON BIG IDEAS (8664 HELD-OUT CGM READINGS, 16 SUBJECTS). MAE AND RMSE IN MG/DL, MARD IN %; r_w: MEDIAN WITHIN-SUBJECT CORRELATION. Δ: PAIRED MAE DIFFERENCE AGAINST THE SUBJECT-MEAN NULL WITH ITS 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL, COMPUTED ON THE READINGS BOTH ROWS PREDICT. THE ENCODER PROBES EXCLUDE 7 OF 28867 READINGS THAT HAVE NO 10-S SEGMENT PASSING THE QUALITY CHECK, WHICH MOVES THE CHRONOLOGICAL CUT SO THAT 4 OF THE 8664 HELD-OUT READINGS DIFFER (PROBE ROWS ARE SCORED ON 8663 READINGS). 1-D CNN, RESIDUAL: MEAN ± STD OVER 3 SEEDS; ALL OTHER ROWS ONE RUN.

	Model	MAE	RMSE	MARD	r	r_w	Δ [95% CI]
null	Subject train mean	15.3	21.6	12.8	0.43	–	–
	Reference-noise floor (MARD 9%)	10.5	–	9.0	–	–	–
	Time of day + subject	15.3	21.8	12.8	0.41	-0.01	+0.0 [-0.5, +0.5]
	Persistence	31.3	44.9	27.3	0.21	–	+16.1 [+3.8, +29.9]
population	Ridge, PPG features	17.3	23.8	14.9	0.05	0.05	+2.0 [+0.7, +3.8]
	1-D CNN	16.9	23.8	14.2	0.21	0.04	+1.7 [+0.5, +3.1]
	2-D CNN (CWT)	17.0	23.8	14.3	0.19	0.03	+1.7 [+0.5, +3.2]
	PointNet (Takens)	17.0	23.8	14.3	0.20	0.03	+1.7 [+0.5, +3.2]
	1D+2D+3D cross-attention	16.7	23.5	14.0	0.25	0.03	+1.4 [+0.2, +2.9]
personal.	Ridge, residual	15.1	21.6	12.7	0.44	0.06	-0.1 [-0.4, +0.1]
	1-D CNN, residual	15.0±0.0	21.9±0.0	12.3±0.0	0.44±0.00	0.04±0.00	-0.2 [-0.7, +0.2]
	Fusion, residual	15.1	21.9	12.3	0.44	0.07	-0.2 [-0.7, +0.3]
probe	PaPaGei-S + ridge	17.2	23.8	14.8	0.07	0.03	+2.0 [+0.6, +3.7]
	Pulse-PPG + ridge	17.0	23.6	14.6	0.15	0.04	+1.7 [+0.4, +3.4]
	AnyPPG + ridge	16.8	23.3	14.4	0.23	0.07	+1.5 [+0.3, +3.2]
	PaPaGei-S, residual	15.1	21.6	12.6	0.43	0.03	-0.2 [-0.5, +0.04]
	Pulse-PPG, residual	15.0	21.5	12.5	0.44	0.07	-0.3 [-0.5, -0.02]
	AnyPPG, residual	15.1	21.4	12.6	0.45	0.17	-0.2 [-0.6, +0.2]

TABLE V

STUBA EPPG, LEAVE-ONE-SUBJECT-OUT, BY CHANNEL; MAE IN MG/DL. NULL: TRAIN MEAN. 1-D CNN: MEAN ± STD OVER 3 SEEDS; OTHER CELLS ONE RUN. PAIRED INTERVALS FOR THE IR1 ROWS ARE GIVEN IN SEC. V-D AND ALL COMPARISONS IN SUPPLEMENTARY SEC. S4.

Channel	Null	Ridge	GBM	1-D CNN
940 nm, site 1 (IR1)	43.5	44.4	48.3	49.7±1.3
940 nm, site 2 (IR2)	43.5	44.9	45.7	–
655 nm, site 1 (R1)	43.5	43.9	49.3	–
655 nm, site 2 (R2)	43.5	44.9	47.7	–
530 nm, site 1 (G1)	43.5	44.1	47.9	–
530 nm, site 2 (G2)	43.5	44.3	46.8	–
six-channel average	43.5	44.1	46.8	–

TABLE VI

CONTROLS THROUGH THE IDENTICAL PIPELINE AND PROTOCOL (1-D CNN; KERMANI–ESMAEILI ROWS MEAN ± STD OVER THREE SEEDS, BIG IDEAS ROW ONE SEED). MAE IN TARGET UNITS; r: PEARSON.

Cohort	Target	Expected	MAE	Null	r
K–E	heart rate (bpm)	recoverable	20.5±0.3	41.9±0.0	0.74±0.01
K–E	age (years)	weak / none	8.9±0.1	8.1±0.0	-0.09±0.01
K–E	glucose	(task)	13.4±0.5	14.7±0.0	0.32±0.02
K–E	glucose, permuted	null	15.6±0.3	14.6±0.0	-0.10±0.13
K–E	glucose, noise input	null	14.6±0.4	14.7±0.0	-0.07±0.09
BIG IDEAs	heart rate (bpm)	recoverable	9.7	14.2	0.63

TABLE VII

REPRESENTATION SENSITIVITY ON BIG IDEAS (GUARDED CHRONOLOGICAL SPLIT, RESIDUAL TARGETS). MAE IN MG/DL; r_w: MEDIAN WITHIN-SUBJECT CORRELATION. ONE WINDOW PER READING AND ONE SEED PER ROW (TABLE IV USES TWO WINDOWS AND THREE SEEDS, HENCE THE SLIGHTLY DIFFERENT DEFAULT ROW); LONGER WINDOWS KEEP ONLY READINGS WITH A GAP-FREE WINDOW OF THAT LENGTH.

Representation	Null	Ridge	r_w	1-D CNN	r_w
5 s, 0.5–8 Hz, z-score (default)	15.3	15.1	0.04	15.1	0.02
10 s	15.3	15.1	0.05	15.1	0.02
30 s (+ HRV features)	15.3	15.1	0.06	15.0	0.04
60 s (+ HRV features)	15.3	15.1	0.07	15.0	0.08
5 s, 0.1–12 Hz	15.3	15.1	0.03	15.1	0.02
5 s, amplitude retained	15.3	15.1	0.07	15.1	0.04
60 s, 0.1–12 Hz, amplitude, HRV	15.3	15.0	0.10	15.0	0.05

5–60-s PPG segment carries about glucose is at most a weak, protocol-dependent cross-subject correlation ($r = 0.32 \pm 0.02$ under leave-one-subject-out, 0.19±0.03 under 5-fold) and, with a personal baseline available, an improvement over that baseline of at most a few tenths of a mg/dL: the 1-D CNN residual arm gives Δ -0.2 mg/dL (95% CI [-0.7, +0.2]), and the best personalised model, a ridge probe on frozen Pulse-PPG embeddings, -0.3 mg/dL ([-0.5, -0.02]), one of only 2 of the 41 paired intervals that lie entirely below zero (the other is the primary Kermani–Esmaeili comparison, -1.4 mg/dL, [-3.0, -0.03]), and by a margin far below the reference's own noise.

The first draft measured the artefact, not the model. With labels collapsed to subject means, a four-subject validation set whose means lie within 104–127 mg/dL cannot produce a large MAE for any reasonable model, and a per-subject oracle MAE of 0.03 mg/dL should have been read as a red flag. We report this plainly because the derivative file is public and carries no provenance, and because subject-wise splitting, the standard remedy, does not detect this class of leak; only a test on the construction of the label column does.

Which leak inflates what. The ladder separates mechanisms that are usually conflated. Window-neighbour leakage (overlapping windows of one recording on both sides) produced the largest inflation here, turning the weak subject-wise correlation ($r = 0.32 \pm 0.02$) into $r = 0.66 \pm 0.05$; identity leakage (recordings separated, subjects shared) helped little on

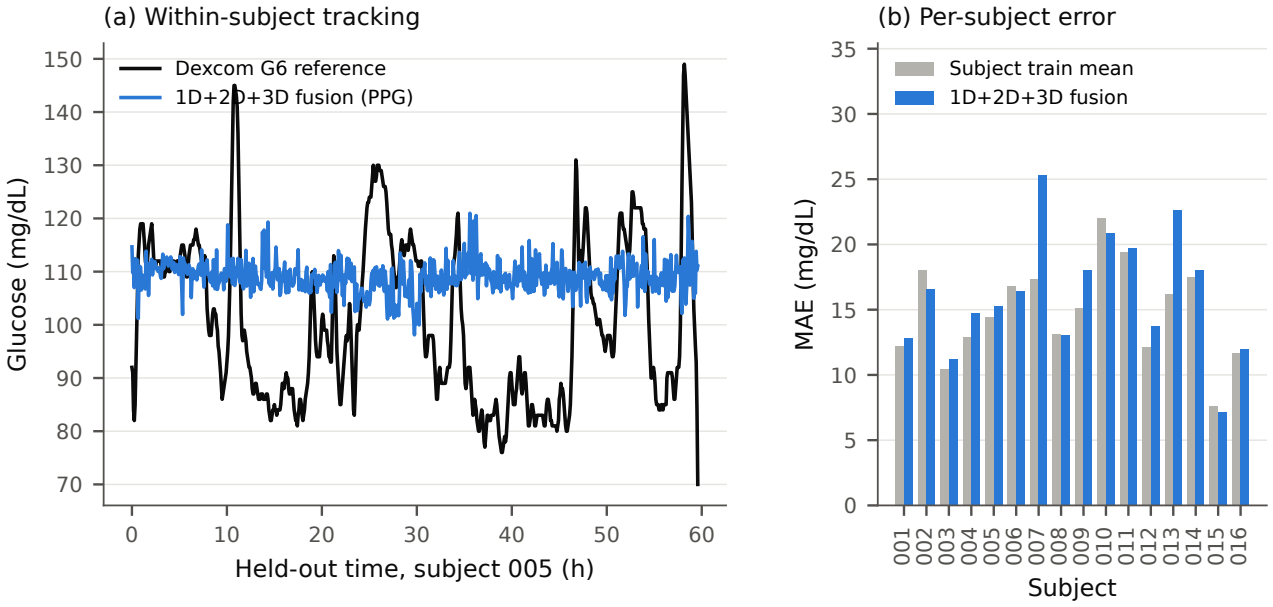


Fig. 5. Within-subject tracking on BIG IDEAs. (a) Dexcom G6 reference and prediction of the population 1D–2D–3D fusion network over the held-out final 30% of one subject's recording. (b) Per-subject MAE of the model against each subject's own training-period mean.

this dataset because most subjects have one or two recordings, and the leaky subject-mean was worse than the train-mean. On the CGM cohort the ladder is flat: a random split over readings lowers the 1-D CNN's MAE by about 0.9 mg/dL (16.1 vs. 16.9) but leaves it above the subject-mean null, and the 1-h guard gap changes the chronological result by less than 0.01 mg/dL at this window length; reading-level leakage is measurable but not conclusion-changing here, and no rung lets any PPG model beat the null. Reports should state which of these their split prevents [22], [27].

Fusion and capacity. Under one shared configuration the 1D–2D–3D fusion network does not consistently improve on its 1-D branch: it is worse on the Kermani–Esmaeili recordings (14.8±0.3 vs. 13.4±0.5) and in the BIG IDEAs residual arm (15.1 vs. 15.0±0.0), marginally better as a BIG IDEAs population model (16.7 vs. 16.9) and on STUBA (47.5 vs. 49.7±1.3), and no paired fusion-versus-1-D interval was computed. It has 6.5× the parameters (548 801 vs. 85 025) and was trained on at most 201 windows per Kermani–Esmaeili fold. We did not tune per model; Sec. V-G gives the reason, so part of that gap may be configuration. We do not conclude that multi-view PPG representations are useless; we conclude that their benefit must be demonstrated against the 1-D branch and a null model under a subject-wise protocol. The multi-view papers in this journal [5], [6] report subject-wise or k-fold results but no null-model row, so the margin over a cohort mean cannot be recovered from them.

Reporting recommendations. (i) Report a null model (train mean, or subject mean, time of day and persistence for within-subject tasks) and a classical feature baseline next to every network, with the reference-noise floor. (ii) Report at the level of independent reference measurements with subject-cluster intervals and a pre-specified primary comparison. (iii) Name the split protocol and the leak it prevents; use a guard gap for time series; ship frozen folds. (iv) Report Clarke or Parkes zone A together with the null model's zone A. (v) Include positive controls. (vi) Test the construction of the label column, not only the split. A TRIPOD+AI checklist and a model information sheet are provided as supplementary material.

Limitations. The fingerstick cohorts are small and, for Kermani–Esmaeili, of unstated diabetes status; BIG IDEAs is prediabetic at most and free-living without signal-quality or accelerometer gating; E4 BVP is a vendor-processed signal. The common representation discards DC amplitude, vasomotion and heart-rate variability, which Sec. V-G only partly restores. Hyper-parameters were shared across models and no model was tuned individually. Pretrained encoders were probed linearly and frozen, and their inputs followed one shared preprocessing rather than each encoder's pretraining pipeline; fine-tuning them on 16–24 subjects would add capacity the cohorts cannot constrain. The benchmark cannot rule out that PPG carries glucose information at scales or in populations it does not cover; it bounds what can be claimed on the public data that exists.

VII. Conclusion

We documented a label-replication artefact in a public PPG-glucose derivative that subject-wise splitting cannot detect, measured a leakage ladder on the same data, and released a null-anchored benchmark harness with machine-checked protocols, frozen folds, controls and clinical anchoring. Under those protocols, on 23+16+24 subjects, no model from ridge regression and gradient boosting on hand-crafted features through a small CNN, re-implemented published regressors and a multi-view fusion network to frozen PPG foundation models improves on the appropriate null by more than a small

fraction of the reference's own noise: the only gains whose intervals exclude zero are -1.4 mg/dL for the pre-specified 1-D CNN on the Kermani–Esmaeili recordings (95% CI [-3.0, -0.03], nominally significant for 1 of 3 seeds) and -0.3 mg/dL for the Pulse-PPG residual probe on BIG IDEAs, against noise floors of 7.1 and 10.5 mg/dL; and none detects a single hypo- or hyperglycaemic reading. The harness is the contribution we expect to outlast the result: any claim of PPG-based glucose estimation on public data can now be run through the same folds, nulls and intervals.

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